BNU

Database Management System

By:

Section:

Company Name: Al-Fatah Online Grocery Store

Submitted to:

Dr. Natash Ali Mian

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Introduction

In today's economy, most of the organisations are recording all their transactions on computer, and so, there should be online systems that provide the service of grocery shopping because it is in high demand as there are loads of transactions carried out every day in a simple grocery store.

Since the provision of web services, it has become a new norm of using applications for using their online services. As a result, people will benefit from online grocery system as they can get their grocery product at any location, and they do not have to make an effort of going to the store.

Company Overview

Since 1941, Al-Fatah is providing a variety of products to its honour customer. Al-Fatah is keen about combining the best of three great words together, namely Quality, Variety, and Customer Service **(Al-Fatah, pp. Welcome to the world of AL-FATAH).**

Existing System Overview

Al Fatah only provides online grocery system on websites. So, we are also providing the online service through application. Moreover, Al-Fatah's existing system only provide the service of cash on delivery, but our system is replacing it by providing online payment.

Proposed System

In our online grocery system, we will be creating a database which will be dealing with the management of the inventory. Other than inventory, online payment and orders will also be handled.

Data Requirements of the Proposed System

User

There are four users of the system, namely Customer, Admin, Sales Manager, and Delivery Man. Each user has a unique user id, user type, contact number, password, first name, last name, and username. Customer has an additional attribute of location.

Category

Category has a unique category id and a category name. Admin adds the category. One admin can add many categories, and one category is only added by one admin.

Subcategory

Subcategory has a unique subcategory id and a subcategory name. Admin adds the subcategory. One admin can add many subcategories, and one subcategory is only added by one admin. Category has subcategory. One category has many subcategories, and one subcategory is present in only one category.

Product

Product has a unique product id , product name, product image, product details, price, issue date, expiry date, quantity and factory code. Admin adds the products. One admin can add many products, and one product is only added by one admin. Subcategory has subcategory. One subcategory has many products, and one product is present in only one subcategory. Customer buys the product. One customer can buy many products, and one product is bought by only one customer. There is also a discount on the product that changes over time. Sales Manager update the quantity of the product. One sales manager can update the quantity of many products, and the quantity of one product is updated by only one sales manager.

Delivery

Delivery has a unique delivery id. Delivery man delivers the delivery. One delivery man delivers many deliveries, and one delivery is only delivered by only one delivery man. Order have information of the delivery. One order has information of only one delivery, and the information of one delivery is only in one order.

Invoice

Invoice has a unique invoice number, total bill, discounted price, and the date on which it was issued. Customer pays the invoice. One customer pays many invoice, and one invoice is paid by only one customer.

Receipt

Receipt has a unique receipt id, receipt number, and receipt date. Invoice generate the receipt. One invoice generates one receipt, and one receipt is only generated by one invoice. Customer receives the receipt. One customer receives many receipt, and one receipt is received by only one customer.

Cart

Cart has a unique cart id, grand total, and subtotal. Product is added to the cart. One product is added to many cart, and one cart adds many products.

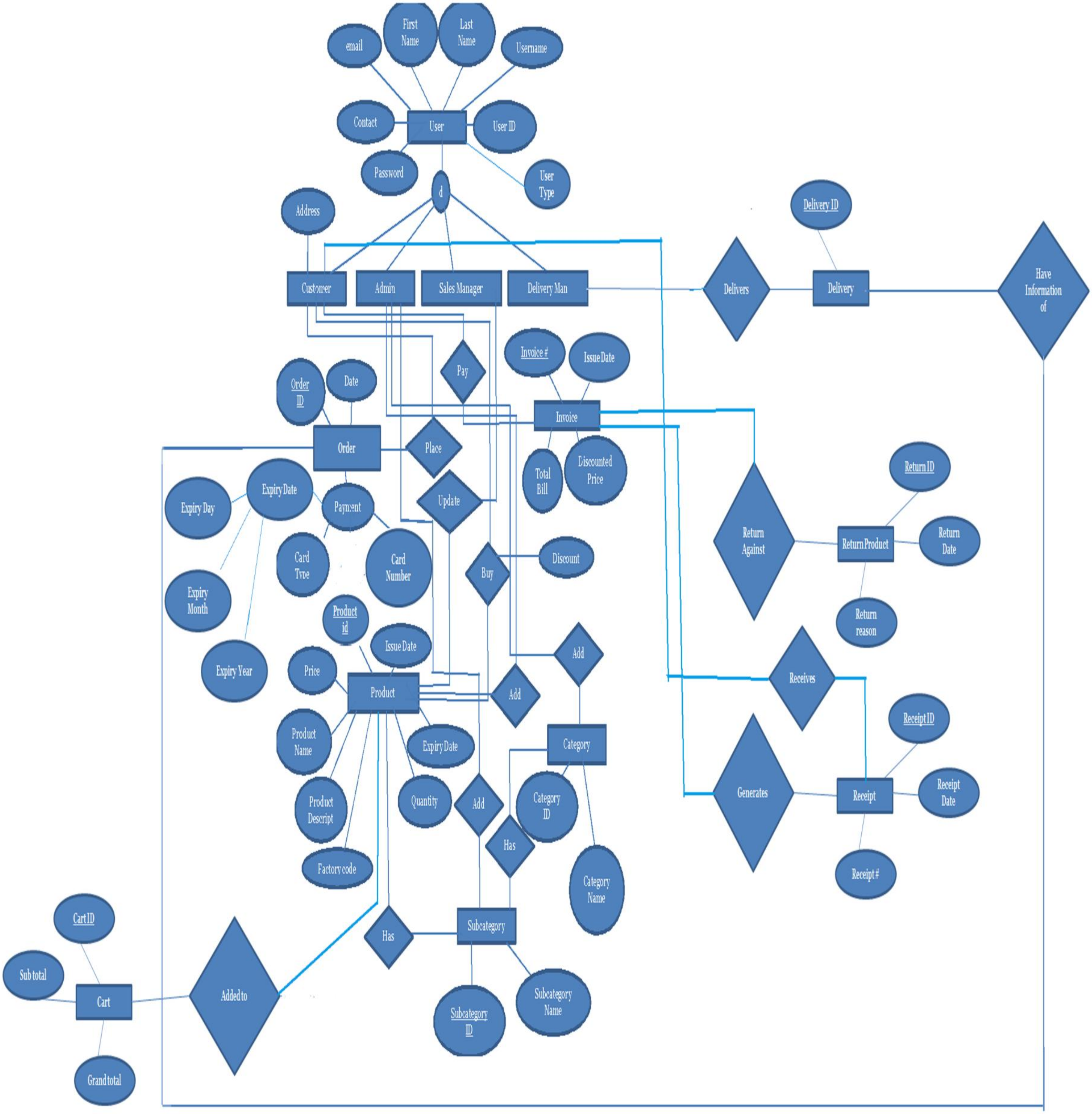
Order

Order has a unique order id, date, and payment, which further extend to card type, CVV, card number, and expiry date (expiry day, expiry month, expiry year). Customer places the order. One customer places many order, and one order is placed by only one customer.

Return Product

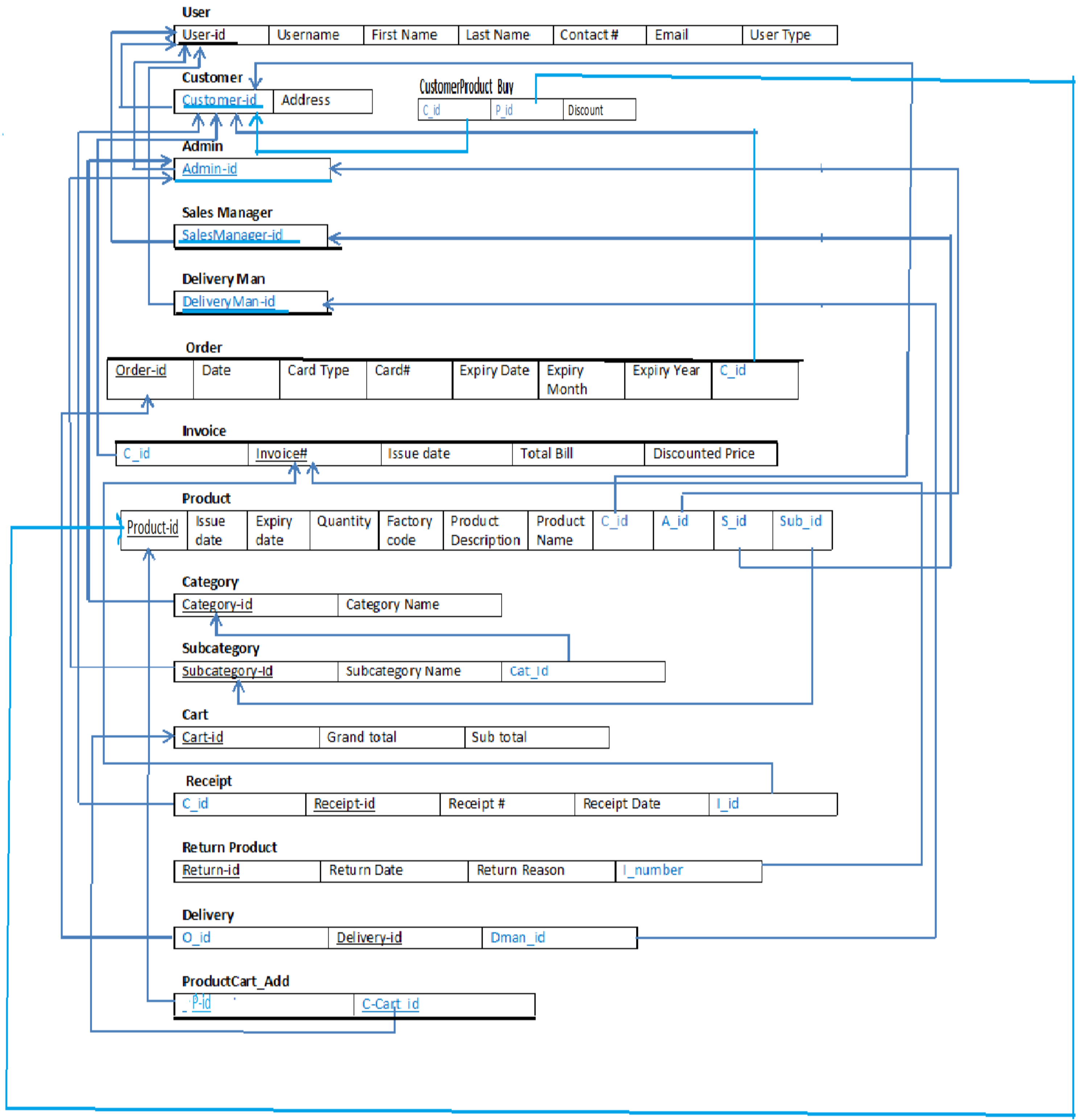
Return product has a unique return id, return date and return reason. Product is returned against invoice. There can be one product returned against one invoice, and one invoice can allow only one product to be returned.

EER Model



email
First Name
Last Name
Username
Contact
User
User ID
Password
d
User Type
Address
Customer
Admin
Sales Manager
Delivery Man
Delivers
Delivery ID
Delivery
Have Information of
Pay
Invoice #
Issue Date
Invoice
Total Bill
Discounted Price
Order ID
Date
Order
Place
Update
Expiry Date
Expiry Day
Payment
Card Type
Card Number
Expiry Month
Expiry Year
Return Against
Return ID
Return Product
Return Date
Return reason
Buy
Discount
Product id
Issue Date
Price
Product
Add
Add
Receives
Receipt ID
Product Name
Expiry Date
Category
Product Descript
Quantity
Add
Category ID
Generates
Receipt
Receipt Date
Has
Factory code
Receipt #
Category Name
Cart ID
Has
Subcategory
Sub total
Added to
Cart
Subcategory Name
Subcategory ID
Grand total

Relational Model



Raw Data

User

Entity Name	Data type	Size	Constraints
User-id	Integer	5	Primary Key
Username	Varchar	15	NOT NULL
First name	Varchar	20	NOT NULL
Last name	Varchar	20	NOT NULL
Contact	Varchar	20	NOT NULL
Email	Varchar	30	NOT NULL
Password	Varchar	30	NOT NULL

Customer

Entity Name	Data type	Size	Constraints	FK Table
Customer-id	Integer	5	Foreign Key+ Primary Key	User
Address	Varchar	50	NOT NULL	

Admin

Entity Name	Data type	Size	Constraints	FK Table
Admin-id	Integer	5	Foreign Key+ Primary Key	User

Sales Manager

Entity Name	Data type	Size	Constraints	FK Table
Sales Manager-id	Integer	5	Foreign Key+ Primary Key	User

Delivery Man

Entity Name	Data type	Size	Constraints	FK Table
Delivery Man-id	Integer	5	Foreign Key+ Primary Key	User

Sub Category

Entity Name	Data type	Size	Constraints	FK Table
Sub Category-id	Integer	5	Primary Key	
Sub Category Name	Varchar	20	NOT NULL	
Admin-id	Integer	5	Foreign Key	Admin
Category-id	Integer	5	Foreign Key	Category

Category

Entity Name	Data type	Size	Constraints	FK Table

Sub Category-id	Integer	5	Primary Key	
Sub Category Name	Varchar	20	NOT NULL	
Admin-id	Integer	5	Foreign Key	Admin

Product

Entity Name	Data type	Size	Constraints	FK Table
Product-Id	Integer	5	Primary Key	
P name	Varchar	20	NOT NULL	
P Details	Varchar	50	NOT NULL	
Price	Integer	6	NOT NULL	
Issue date	Varchar	10	NOT NULL	
Expire date	Varchar	10	NOT NULL	
Quantity	Varchar	10	NOT NULL	
Factory Code	Integer	10	NOT NULL	
Customer-id	Integer	5	Foreign key	Customer
Admin-id	integer	5	Foreign key	Admin
Sales Manager - id	Integer	5	Foreign key	Sales Manager
Sub Category- id	Integer	5	Foreign key	Subcategory

Delivery

Entity Name	Data type	Size	Constraints	PK Table
Delivery-id	Integer	5	Primary Key	
Order – id	Integer	5	Foreign Key	Order
Delivery Man-id	Integer	5	Foreign Key	Delivery Man

Invoice

Entity Name	Data type	Size	Constraints	PK Table
Invoice No	Integer	5	Primary Key	
Discounted Price	Integer	5	NOT NULL	
Total Bill	Integer	7	NOT NULL	
Invoice Issue Date	Varchar	10	NOT NULL	
Customer-id	Integer	5	Foreign Key	Customer

Receipt

Entity Name	Data type	Size	Constraints	PK Table
Receipt-id	Integer	5	Primary Key	
Receipt No	Integer	5	NOT NULL	
Receipt Date	Varchar	10	NOT NULL	
Customer – id	Integer	5	Foreign Key	Customer
Invoice No	Integer	5	Foreign Key	Invoice

Cart

Entity Name	Data type	Size	Constraints	PK Table

Cart id	Integer	5	Primary Key	
Number of products	Integer	5	NOT NULL	
Sub Total	Integer	5	NOT NULL	
Grand Total	Integer	5	NOT NULL	

Order

Entity Name	Data type	Size	Constraints	PK Table
Order – id	Integer	5	Primary Key	
Order Date	Varchar	10	NOT NULL	
Card type	Varchar	10	NOT NULL	
CVV	Integer	3	NOT NULL	
Expiry month	Integer	2	NOT NULL	
Expiry year	Integer	4	NOT NULL	
Customer –id	Integer	5	Foreign key	Customer
Card Number	varchar	20	NOT NULL	

Return Product

Entity Name	Data type	Size	Constraints	PK Table
Return no	Integer	5	Primary Key	
Return date	Varchar	10	NOT NULL	
Return reason	Varchar	150	NOT NULL	
Invoice – id	Integer	5	Foreign key	Invoice

Buys

Entity Name	Data type	Size	Constraints	PK Table
Customer -id	Integer	5	Primary Key+ Foreign key	Customer
Product-id	Integer	5	Primary key + Foreign Key	Product
Percentage of discount	Integer	3	NOT NULL	

Product_Cart_Add

Entity Name	Data type	Size	Constraints	PK Table
Product-id	Integer	5	Primary Key+ Foreign key	Product
Cart-id	Integer	5	Primary key + Foreign Key	Cart

SQL Statement

1

DBMS_project

Limit 100 Run

```
--How many products are added by admin_id 5 ?

select *
from Product
where admin_id = 5;
```

Success
10 rows

Explore SQL Data Chart Export

11:30 PM
0.171 seconds

Product_id	Pname	Pdetails	price	issuedate	expiredate	quantity	factorycode	customer_id	admin_id	salesmanage
2	Blue Band	Blue Band Margarine Spread 200 gm	140	29/Sept/2019	29/Sept/2020	1000	J628W2	2	5	8
4	Adam Buffalo	Adam Buffalo Mozzarella Cheese 2 k	2445	29/Dec/2019	29/Dec/2020	1000	Q628W1	2	5	8
6	Adam Pizza Che	Adam Pizza Cheese 227 gm	385	29/Feb/2019	29/Feb/2020	1000	Q628W8	2	5	9
8	Golden Eggs	Farm Fresh Golden Eggs 12 Pcs Pac	180	29/Sept/2019	29/Sept/2020	1000	E6G8W2	2	5	8
10	Cake Rusk	Aroma Bran Cake Rusk	799	29/Dec/2019	29/Dec/2020	1000	A628W3	2	5	8
12	Rusk	Aroma Garlic Rusk	750	29/Feb/2019	29/Feb/2020	1000	A628W1	2	5	8
14	Flakes Cereal	Fauji Corn Flakes Cereal 250 gm	240	29/Sept/2019	29/Sept/2020	1000	B62RW2	2	5	8
20	Horlickst	Horlicks Drinking Powder Chocolate	535	29/Sept/2019	29/Sept/2020	1000	S628W2	2	5	8
22	Gatorade	Gatorade Sports Drink Lemon Lime 5	195	29/Dec/2019	29/Dec/2020	1000	K6R8W3	2	5	8
26	7 Up Tin	7 Up Soft Drink Tin 300 ml	100	29/Sept/2019	29/Sept/2020	1000	V6G8W2	2	5	8

2

DBMS_project

```
--How many products are added by which admin?

select count(product_id),admin_id
from product
GROUP BY admin_id;
```

Success
3 rows

Explore SQL Data Chart Export

count(product_id)	admin_id
7	4
10	5
10	6

3

DBMS_project

```
--Find the Products brought NOT by customer id 1 & 2 by showing their name and price

select Pname,price,customer_id
from product
where customer_id NOT IN (1,3);
```

Success

10 rows

Explore | SQL | Data | Chart | Expo

Pname	price	customer_id
Blue Band	140	2
Adam Buffalo	2445	2
Adam Pizza Che	385	2
Golden Eggs	180	2
Cake Rusk	799	2
Rusk	750	2
Flakes Cereal	240	2
Horlickst	535	2
Gatorade	195	2
7 Up Tin	100	2

4

DBMS_project

Limit 100 | Run

```
--Find the Product who has first word adam ;

select *
from product
where pname like "adam%";
```

Success

4 rows

Explore | SQL | Data | Chart | Export

12:30 AM

0.157 seconds

Product_id	Pname	Pdetails	price	issuedate	expiredate	quantity	factorycode	customer_id	admin_id	salesman
1	Adam Butter	Adam Butter 100 gm	140	29/Dec/2019	29/Dec/2020	1000	J628W3	1	4	7
4	Adam Buffalo	Adam Buffalo Mozzarella Cheese 2 I	2445	29/Dec/2019	29/Dec/2020	1000	Q628W1	2	5	8
5	Adam Cheddar	Adam Cheddar Cheese 2 kg	1795	29/Sept/2019	29/Sept/2020	1000	Q628W2	3	6	7
6	Adam Pizza Cheese	Adam Pizza Cheese 227 gm	385	29/Feb/2019	29/Feb/2020	1000	Q628W8	2	5	9

5

DBMS_project

Limit 100 Run

```sql
--Find the Product who has word is letter Eggs ;

select *
from product
where pname like "%Eggs";
```

Success
2 rows

Explore | SQL | Data | Chart | Export |

12:43 A
0.163 second

Product_id	Pname	Pdetails	price	issuedate	expiredate	quantity	factorycode	customer_id	admin_id	salesman
7	Classic Eggs	Farm Fresh Classic Eggs 12 Pcs Pac	150	29/Dec/2019	29/Dec/2020	1000	E6G8W3	3	6	7
9	Omega 3 Eggs	Farm Fresh Omega 3 Eggs 12 Pcs P	205	29/Feb/2019	29/Feb/2020	1000	E6G8W1	3	6	9

6

DBMS_project

Limit 100 Run

```sql
--Find the Product who has word letters dam and dont know the first letter and rest of words ;

select *
from product
where pname like "_dam%";
```

Success
4 rows

Explore | SQL | Data | Chart | Export |

12:44 AM
0.111 seconds

Product_id	Pname	Pdetails	price	issuedate	expiredate	quantity	factorycode	customer_id	admin_id	salesman
1	Adam Butter	Adam Butter 100 gm	140	29/Dec/2019	29/Dec/2020	1000	J628W3	1	4	7
4	Adam Buffalo	Adam Buffalo Mozzarella Cheese 2 I	2445	29/Dec/2019	29/Dec/2020	1000	Q628W1	2	5	8
5	Adam Cheddar	Adam Cheddar Cheese 2 kg	1795	29/Sept/2019	29/Sept/2020	1000	Q628W2	3	6	7
6	Adam Pizza Cheese	Adam Pizza Cheese 227 gm	385	29/Feb/2019	29/Feb/2020	1000	Q628W8	2	5	9

7

DBMS_project

```
-- what are the total sales of today?


Select sum(totalbill)
from invoice;
```

Success

1 rows

sum(totalbill)
10800

8

DBMS_project

```sql
-- which products are going to expire between 1/Dec/2019 to 31/dec/2019?


Select pname
from product
where expiredate BETWEEN '01/Dec/2020' AND '31/Dec/2020';
```

Success

27 rows

Explore | SQL | Data | Chart

pname
Adam Butter
Blue Band
Lurpak Butter
Adam Buffalo
Adam Cheddar
Adam Pizza Cheese
Classic Eggs
Golden Eggs
Omega 3 Eggs
Cake Rusk
Aroma BreadAroma
Rusk
Porridge Cereal
Flakes Cereal

9

DBMS_project

```
-- How many customers are Registered?



Select *
from users
where usertype ='Customer';
```

Success

3 rows

Explore | SQL | Data | Chart | Export

userid	Username	Firstname	Lastname	Email	Password_s	contact	usertype
1	Hassan1	Hassan	Butt	Hassan7196@yahoo.com	Hassan12	0315-0781908	Customer
2	Talal1	Talal	Amir	Talal7196@yahoo.com	talal12	0315-0781909	Customer
3	Raza1	Raza	Ahmed	Raza7196@yahoo.com	Raza12	0315-0781910	Customer

10

DBMS_project

```
-- what are the SubCategory of Dairy?


Select subcategoryName
from subcategory
where category_id =1;
```

Success

3 rows

Explore | SQl

subcategoryName
Butter & Margarine
Cheese & Cream
Eggs

11

DBMS_project

```sql
-- what are the address of the registered customer ?(check for delivery Purposes)


select users.userid,users.Firstname,users.Lastname,customer.address
from users
INNER JOIN customer ON userid = customer_id;
```

Success

3 rows

Explore | SQL | Data | C

userid	Firstname	Lastname	address
1	Hassan	Butt	DHA Phase 6, Lahore, Punjab 54810
2	Talal	Amir	DHA Phase 1, Lahore, Punjab 64310
3	Raza	Ahmed	DHA Phase 2, Lahore, Punjab 64310

12

DBMS_project

```
-- Display all the invoices generated along with customer's name.



select invoice.InvoiceNo,
users.Firstname,
users.lastname,
invoice.TotalBill
from (invoice
INNER JOIN users on users.userid = invoice.customer_id);
```

Success

3 rows

Explore SQL

InvoiceNo	Firstname	lastname	TotalBill
1	Hassan	Butt	1000
2	Talal	Amir	800
3	Raza	Ahmed	9000

13

DBMS_project

```
-- which admin updated which product?



select
Users.Firstname,
Users.Lastname,
product.Pname
from Users
INNER JOIN product ON product.admin_id = users.userid
ORDER BY users.Firstname,Users.Lastname;
```

Success

27 rows

Explore

Firstname	Lastname	Pname
Hassan	Butt	Flakes Cereal
Hassan	Butt	Adam Buffalo
Hassan	Butt	Gatorade
Hassan	Butt	Golden Eggs
Hassan	Butt	7 Up Tin
Hassan	Butt	Rusk
Hassan	Butt	Blue Band
Hassan	Butt	Horlickst
Hassan	Butt	Adam Pizza Cheese
Hassan	Butt	Cake Rusk
Jamil	Jutt	Johar Joshanda
Jamil	Jutt	7 Up
Jamil	Jutt	Ispaghol
Jamil	Jutt	Adam Butter

Contribution

We have successfully created the online grocery shopping system. We hope that this system will help the community as they do not have to travel a lot for buying products in grocery store. Instead, they can order their products at their location. Moreover, for people who are the frequent users of web applications instead of websites, We are also going to provide the grocery service through application as well so that maximum of people can get a chance to use our service.

Experience

It was a great experience regarding creating a database for an online grocery shopping store. We have learned to work on soft wares like My SQL Server and Pop SQL. Moreover, we have also learned that how an online grocery store works i.e. what things the system requires for storing it in a database, and so, we have created the database accordingly.

Conclusion

We look forward to create more online systems in future and to work with new softwares as well.